Biliary Atresia, Part Eight: Prognosis after the Kasai Operation. Clearance of Jaundice and Long-Term Survival with the Native Liver.

Picture a baby girl who had a Kasai operation for biliary atresia twelve weeks ago. Her stools are yellow, she is gaining a little weight, and this morning her total bilirubin came back at one point six milligrams per decilitre. Her parents have read a great deal on the internet, and they ask the question every family asks at this visit. Did the operation work? Will she keep her own liver, and if so, for how long? Answering them well requires two things. You need to know which early numbers actually predict the future, and you need an honest picture of what happens to children with biliary atresia over twenty and thirty years.

This chapter is built around one main idea. The bilirubin level in the first three months after the Kasai is the most useful early sign of how the operation has gone. An infant whose total bilirubin falls below two milligrams per decilitre within three months has roughly a four in five chance of being alive without a transplant two years later. An infant whose bilirubin stays above that level has only about a one in five chance. But clearing the jaundice is not the same as being cured. In large national studies, about half of all children keep their own liver to five years, and only about a quarter keep it to twenty years. Almost every child who does keep the native liver has some degree of chronic liver disease, and in the long run it is portal hypertension, recurrent cholangitis and nutrition that decide whether that liver lasts. Prognosis is therefore not a single verdict given at three months. It is an estimate that you revise at every visit, for the rest of the patient's life.

A few terms first. The Kasai portoenterostomy is the operation in which the surgeon removes the scarred remnant of the bile duct at the porta hepatis, the point where the ducts leave the liver, and sews a loop of small bowel onto the raw surface so that bile can drain from the tiny surviving ductules into the gut. Clearance of jaundice means that the bilirubin falls to normal or nearly normal after the operation. Native-liver survival means being alive with one's own liver, without a transplant; it is the main outcome in every study discussed here. Overall survival includes children who are alive after a transplant. Portal hypertension is raised pressure in the portal vein, the vessel that carries blood from the gut to the liver; it shows itself as a large spleen, a low platelet count, varices, which are dilated veins in the oesophagus that can bleed, and ascites, which is fluid in the abdomen. Cholangitis is bacterial infection of the bile ducts.

The evidence for this chapter comes from several kinds of source. There is the prospective North American cohort of the Childhood Liver Disease Research Network, known as ChiLDReN, reported by Shneider and colleagues in the Journal of Pediatrics in 2016. There are national cohorts from France, reported by Serinet and colleagues in Pediatrics in 2009 and by Fanna and colleagues in the Journal of Pediatric Gastroenterology and Nutrition in 2019; the Japanese Biliary Atresia Registry, summarised by Nio in 2017 and analysed for portal hypertension by Okubo and colleagues in 2025; and national series from the Netherlands, Canada, Switzerland, the Nordic countries and Taiwan. There are studies that follow children into adulthood, from King's College Hospital in London, Paris and Japan. And there are the documents that guide practice: the 2014 guideline on evaluating children for liver transplantation from the American Association for the Study of Liver Diseases, the 2017 cholestasis guideline from the

North American and European paediatric gastroenterology societies, and the 2021 guideline of the Japanese Biliary Atresia Society.

Part one. Why drainage predicts, and why it is not enough.

To understand prognosis after the Kasai, it helps to think of two clocks. The first clock is drainage. The Kasai can only work if the porta still contains ductules that are large enough and numerous enough to carry bile. The older the infant and the more advanced the fibrosis at the time of surgery, the fewer of these ductules remain. If bile does flow, the bilirubin falls over a period of weeks. The bilirubin level at around three months therefore summarises several things at once: how many ductules were patent, whether early cholangitis damaged them, and how much fibrosis was already present. This first clock is set within weeks of the operation, and it is read from the bilirubin.

The second clock is fibrosis. Biliary atresia is not only a disease of the large duct outside the liver. It is also an inflammatory, scarring disease of the small ducts inside the liver, and that process does not stop when jaundice clears. Episodes of cholangitis add further injury, and sluggish bile flow adds more. Over months and years, fibrosis raises the pressure in the portal vein. The spleen enlarges, the platelet count falls, varices form and, in some children, ascites appears. Eventually, some livers fail. This second clock keeps running in children who look perfectly well and have a normal bilirubin. It is read from platelets, spleen size, albumin, liver stiffness and, later, from varices.

The practical meaning of the two-clock model is simple. Survival in the first two years depends mostly on the first clock: did the Kasai drain? Survival into adolescence and adulthood depends mostly on the second clock: how fast is the fibrosis progressing? A child can win on the first clock and still lose on the second.

Part two. What counts as clearance.

Before comparing numbers between centres, it is worth knowing that clearance of jaundice is not defined the same way everywhere. The North American ChiLDReN network used a total bilirubin below two milligrams per decilitre, which is about thirty-four micromoles per litre, reached at any point within the first three months after surgery. The American transplant guideline uses the same value. The Taiwanese national programme called a child jaundice-free if the bilirubin was below two milligrams per decilitre at three months. British, Nordic and Finnish series usually use a stricter threshold, a bilirubin below twenty micromoles per litre, which is about one point two milligrams per decilitre, reached within six months. The French national cohort used a bilirubin of twenty micromoles per litre or less at any time after the operation. And the American START trial of steroids defined successful bile drainage as a bilirubin below one point five milligrams per decilitre with the native liver still in place at six months.

These definitions are not interchangeable. A stricter threshold measured later will always produce a lower rate than a looser threshold measured earlier. So when you compare your own unit's results with a published series, check three things: the threshold, the time point, and who is in the denominator.

Part three. The three-month bilirubin.

The key study is the ChiLDReN analysis by Shneider and colleagues. They followed one hundred and thirty-seven infants who had a Kasai between 2004 and 2011 in North American centres. Sixty-eight of them, exactly half, reached a total bilirubin below two milligrams per decilitre within three months. The difference in outcome was dramatic. Two years after surgery, eighty-six percent of the infants who had reached that threshold were alive without a transplant, compared with only twenty percent of those who had not.

The same study showed that infants whose bilirubin stayed at two or above were far more likely to develop the complications of failing liver function. Compared with infants who cleared, their odds of ascites were about six times higher, their odds of a low albumin about seven and a half times higher, and their odds of coagulopathy, meaning a prolonged clotting time, almost eleven times higher. Their odds of needing a liver transplant were about twelve times higher, and their odds of transplant or death about seventeen times higher. Their weight gain was also clearly worse. In other words, the three-month bilirubin sorts infants into two very different futures, and it does so early enough for you to act: to intensify nutrition, and to involve the transplant team before the child deteriorates.

Several other early markers add to the bilirubin. A study from Sydney and Toronto by Nightingale and colleagues, published in 2017, looked at two hundred and seventeen infants. At three months after surgery, the bilirubin, the albumin and the treating centre all independently predicted whether the child would have the native liver at two years. A bilirubin below seventy-four micromoles per litre, which is about four point three milligrams per decilitre, predicted native-liver survival very well, with an area under the curve of about zero point nine. An albumin above thirty-five grams per litre was also useful, with an area under the curve of about zero point seven six. Importantly, a low albumin was a bad sign even in infants who were no longer jaundiced.

A Finnish study from 2025 by Anderson, Hukkinen and colleagues added serum bile acids. In infants who went on to clear their jaundice, the bilirubin at one month did not predict their later outcome, but the bile acid level at one month did, with an area under the curve of about zero point eight two. The same study showed that "normal" bilirubin is really a range. Among children who had cleared their jaundice, those whose bilirubin was still above eight and a half micromoles per litre at six months had about thirteen times the risk of developing portal hypertension. The low end of normal matters.

Liver stiffness, measured with ultrasound elastography, also helps. In a Korean study of sixty-nine children by Hahn and colleagues in 2013, a stiffness above about twenty kilopascals three months after the Kasai predicted later liver-related events with excellent accuracy. And at the other end of childhood, the King's College group found that a bilirubin of twenty-one micromoles per litre or more at the age of sixteen predicted the need for a transplant later in adulthood, with an area under the curve of about zero point eight five. Put simply, bilirubin tells you whether the Kasai drained. Albumin, bile acids and stiffness tell you how much liver damage remains despite drainage.

Part four. How often jaundice clears, and how long the liver lasts.

Across published programmes, the proportion of infants who clear their jaundice ranges from about one third to about three quarters. A small recent Finnish series reported seventy-five percent clearance by six months. The Nordic countries together reported normalisation of bilirubin in sixty-four percent of one hundred and forty-eight infants operated between 2005 and 2016. In the START trial, bile drainage

at six months was achieved in fifty-nine percent of infants who received steroids and forty-nine percent of those who received placebo, a difference that was not statistically significant. The ChiLDReN cohort, as we have seen, reported fifty percent. The French national cohort, which covered three decades and more than thirteen hundred operations, reported clearance in thirty-nine percent.

Taiwan provides the most instructive comparison. Before the national stool colour card programme, only about thirty-five percent of infants were jaundice-free at three months. After universal stool colour cards were introduced, the proportion operated before sixty days of age rose from about forty-nine to about sixty-six percent, and the proportion jaundice-free at three months rose to about sixty-one percent. Earlier surgery meant more clearance.

What about the long run? The largest population data come from France. Fanna and colleagues described more than fourteen hundred children born between 1986 and 2015, of whom thirteen hundred and forty had a Kasai. Native-liver survival was forty-one percent at five years, thirty-five percent at ten years, twenty-six percent at twenty years and twenty-two percent at thirty years. These figures were remarkably stable across the decades. More than half of the children eventually had a transplant, and overall survival was eighty-one percent, because transplantation rescues most children whose Kasai fails; in France, survival after transplant was seventy-nine percent at twenty-eight years. An earlier French analysis by Serinet and colleagues gave similar numbers: about fifty-seven percent at two years, thirty-eight at five, thirty-two at ten and twenty-nine at fifteen.

The Japanese registry, which has followed more than three thousand children since 1989, reports somewhat better long-term figures, with native-liver survival of about forty-nine percent at twenty years and overall survival of about eighty-nine percent at twenty years. Other national series fall in a similar range. In the Netherlands, native-liver survival at four years was forty-six percent. In Canada it was between thirty-five and thirty-nine percent at four years. In Switzerland it was about thirty-seven percent at four years. In the Nordic countries it was fifty-three percent at five years. In Taiwan, five-year jaundice-free survival with the native liver rose from about twenty-seven percent before stool card screening to about sixty-four percent after it. A reasonable benchmark, then, is that a good programme keeps about half of its patients on their own liver at four to five years. Interestingly, in the Netherlands, centres with small caseloads did not do worse, so case volume is not the whole story.

Part five. What moves the odds.

The first modifier is age at the Kasai. In France, native-liver survival at twenty-five years was thirty-eight percent when the Kasai was done in the first month of life, twenty-seven percent in the second month, twenty-two percent in the third, and nineteen percent when it was done later. In Canada, native-liver survival at four years was forty-nine percent when the operation was done at thirty days or younger, thirty-six percent between thirty-one and ninety days, and twenty-three percent after ninety days. There is no sharp cut-off; the odds fall steadily with every week. But late surgery is not futile. In France, about one in five children operated after three months still had their own liver at twenty-five years. Age sets the starting odds, and the three-month bilirubin then updates them for the individual child.

The second modifier is the experience of the centre. A British and Irish study from the mid-1990s by McKiernan and colleagues found better transplant-free survival in centres that performed more

operations, and that finding helped drive the centralisation of biliary atresia surgery in England and Wales. In France, however, caseload mattered only in the earlier era, and in the Netherlands it did not seem to matter at all.

The third modifier is the type of biliary atresia. A meta-analysis found that infants with cytomegalovirus-associated biliary atresia, identified by a positive cytomegalovirus immunoglobulin M test, were less likely to clear their jaundice, with an odds ratio of about zero point five, although the difference in two-year survival did not reach significance. Syndromic biliary atresia, the form associated with splenic malformations, and cystic biliary atresia are often described as having worse and better outcomes respectively, but the size of these effects varies between series, and you should be careful not to overstate them to families.

The fourth modifier is cholangitis, which was the subject of the previous chapter. A Taiwanese study published in 2026 found that among children who had cleared their jaundice, three or more episodes of cholangitis a year were associated with a poorer outcome, with an odds ratio of about two and a half, and a direct bilirubin of one point one milligrams per decilitre or more at three months carried an odds ratio of about eight.

The fifth, and in the long run the most powerful, modifier is portal hypertension. Okubo and colleagues analysed three thousand seven hundred and seventy-seven patients from the Japanese registry. Nearly half of them, about forty-six percent, developed portal hypertension, and so did about half of those who were still alive with their own liver at twenty years. For children who never developed portal hypertension, native-liver survival was about ninety percent at ten years and still about eighty-eight percent at thirty years. For children with portal hypertension that did not need treatment, it was about eighty-one percent at ten years and sixty-eight percent at thirty years. And for children whose portal hypertension needed treatment, it was about sixty-two percent at ten years and only about forty-six percent at thirty years. In plain terms, without portal hypertension a native liver usually lasts; once portal hypertension needs treatment, about half of these children will lose the liver within three decades, and you should plan for transplant.

Part six. A native liver is rarely a normal liver.

Families often hear that the operation "worked" and assume their child is now healthy. The ChiLDReN network studied two hundred and nineteen children who were alive with their own liver five or more years after the Kasai, at a median age of almost ten years. Their findings, reported by Ng and colleagues in the Journal of Pediatrics in 2014, are sobering. More than ninety-eight percent had clinical or laboratory evidence of chronic liver disease. Sixty-two percent had had at least one episode of cholangitis. Fifty-six percent had a large spleen and forty-three percent had a low platelet count. Seventeen percent had had ascites, fifteen percent had had a bone fracture, and nine percent had had bleeding from varices. Only about half had a normal health-related quality of life. A systematic review of twenty-year survivors by Bijl and colleagues in 2013 found a similar picture, with cholangitis in about sixty percent and portal hypertension in about half.

Adolescence and young adulthood bring a second period of risk. King's College Hospital followed three hundred and ninety-seven children who had a Kasai between 1980 and 1996. One hundred and eleven of them, twenty-eight percent, reached the age of sixteen with their own liver. Of eighty-nine

who could be followed further, twenty-two needed a transplant after the age of sixteen. The risk was higher in those who had had portal hypertension before sixteen, about seven times higher; in those with varices, about eight and a half times higher; and in those who had cholangitis during adolescence, about five times higher. In Paris, sixty-three patients were alive with their own liver more than twenty years after the Kasai, with complications still appearing in adulthood. And in a Japanese series of twenty-two adults with a median age of twenty-five, twenty-one had portal hypertension, half had cholangitis and jaundice, and four needed a transplant between the ages of twenty and twenty-seven. Transition to adult care must therefore carry the bilirubin, platelet and variceal history with it, and the adult team should know that liver failure, hepatopulmonary syndrome and, rarely, liver cancer can still appear decades after the operation.

Part seven. When to change the plan.

The ChiLDReN investigators, in a review by Sundaram and colleagues in Liver Transplantation in 2017, summarised the reasons a child with biliary atresia comes to transplant. The first is a failed Kasai, meaning a bilirubin that does not fall. The second is malnutrition and growth failure despite optimised calories, medium-chain triglycerides and fat-soluble vitamins. The third is recurrent or intractable cholangitis, especially when there are bile lakes. The fourth is the complications of portal hypertension: bleeding that cannot be controlled, ascites and hypersplenism. The fifth is pulmonary: hepatopulmonary syndrome, in which dilated vessels in the lungs cause low oxygen levels, or portopulmonary hypertension, in which the pulmonary arteries narrow. These two can appear whatever the bilirubin.

On timing, the 2014 American guideline on transplant evaluation advises considering referral once the bilirubin stays at or above two milligrams per decilitre after the Kasai, and referring promptly when it stays above six milligrams per decilitre more than three months after the operation. Referral is not the same as listing. The point is to start evaluation, nutritional rescue and, where possible, a living-donor work-up before the child decompensates.

A practical monitoring frame follows from the evidence, although it is not a guideline schedule. At one month, check the bilirubin, and if available the bile acids, the stool colour and the weight; rising bilirubin or paler stools should be treated as cholangitis or as failure of drainage. At three months, record the total bilirubin, the albumin, the clotting and the growth; a bilirubin of two or more means planning a referral, and above six means referring now. At six months, see whether the bilirubin is truly normal, and look at platelets, spleen size and, if possible, stiffness. At every visit thereafter, look at platelets, spleen, albumin, growth and vitamins, because new thrombocytopenia or a newly palpable spleen calls for a work-up for portal hypertension. Every year, check oxygen saturation to screen for hepatopulmonary syndrome. And in adolescence, a bilirubin of twenty-one micromoles per litre or more, or the presence of varices, should place the young person on the adult transplant pathway. Above all, write the three-month bilirubin into every clinic letter. It is the single number that best predicts the next two years.

Part eight. What the guidelines say.

When you compare the guidelines, the striking finding is how little they say about prognosis. The 2014 guideline of the American Association for the Study of Liver Diseases on transplant evaluation is the

only society document that gives a bilirubin trigger for referral, the thresholds of two and six milligrams per decilitre already described. The European Association for the Study of the Liver has no guideline on biliary atresia. The 2017 cholestasis guideline of the North American and European paediatric gastroenterology societies is a diagnostic document; it notes that native-liver survival falls with age at the Kasai, but it does not address clearance, monitoring or referral. The 2021 guideline of the Japanese Biliary Atresia Society answered twenty-one of its twenty-five clinical questions, but it explicitly declined to make recommendations about liver transplantation because the evidence was insufficient. So on the definition of clearance the documents differ; on the referral trigger there is a material gap, with one society giving a number and the others silent; on the benefit of early surgery they agree; and on lifelong follow-up none of them gives explicit guidance. In practice, prognosis after the Kasai rests on cohorts and registries rather than on guidelines.

Part nine. Traps, pearls and talking to families.

There are several common traps. The first is to think that a normal bilirubin means the child is cured. More than ninety-eight percent of native-liver survivors have chronic liver disease, so keep watching the platelets and the spleen. The second is to judge success from stool colour alone. Stools can look pigmented even when the bilirubin is above two, so measure rather than infer. The third is to believe that steroids improved clearance. In the START trial, bile drainage was fifty-nine percent with steroids and forty-nine percent with placebo, which was not significant, and survival with the native liver at two years was the same in both groups. The fourth is to assume that a late Kasai is pointless. In France, about one in five children operated after three months still had their own liver at twenty-five years, so the decision should be made case by case. The fifth is to conclude that your unit is doing badly because its numbers are lower than a published series; first check that you are comparing the same definition and the same time point. And finally, a child of six with a normal bilirubin and a new low platelet count has not suddenly developed liver disease. The second clock was running all along.

How should you talk to families at three months? If the bilirubin is below two milligrams per decilitre, you can say that about eighty-six percent of children like theirs are still free of transplant two years later, but that this is a lifelong condition and cholangitis and portal hypertension can still occur. If the bilirubin is between two and six, explain that drainage is partial, that nutrition needs to be pushed now, and that you would like the transplant team to meet them early. If it is above six, explain that the Kasai has not worked well enough, that you are referring promptly, and that most such children need a transplant in the first two years. And tell every family that overall survival, including transplant, is above eighty percent, that pale stools or fever must be reported the same day, and that you will plan together for adolescence. Avoid words like "worked" and "failed" as final verdicts. Describe what the bilirubin means for the next two years and what you will watch afterwards.

To pull it together, here are the numbers worth carrying into clinic. A bilirubin below two milligrams per decilitre within three months means a two-year transplant-free survival of about eighty-six percent, compared with twenty percent if it stays higher, and about half of infants reach it. Albumin above thirty-five grams per litre, low bile acids and low liver stiffness add useful information. Definitions of clearance differ, so compare like with like. In France, native-liver survival is about forty-one percent at five years, twenty-six percent at twenty and twenty-two percent at thirty; in Japan it is about forty-nine percent at twenty years. Overall survival is above eighty percent because transplantation rescues failed

operations. Age at the Kasai matters across the whole range, from about thirty-eight percent long-term native-liver survival when operated in the first month to about nineteen percent when operated after the third. In Japan, thirty-year native-liver survival is about eighty-eight percent without portal hypertension and about forty-six percent when portal hypertension needs treatment. More than ninety-eight percent of five-year native-liver survivors have chronic liver disease. At King's, only twenty-eight percent reached sixteen with their own liver, and a quarter of those needed a transplant later. And the American guideline advises prompt referral when the bilirubin stays above six milligrams per decilitre beyond three months, with referral considered from two.

What should you do differently in clinic? Record the three-month bilirubin and the albumin in every letter and use them to set the plan. Involve the transplant team early for any infant whose bilirubin stays at two or above, and refer without delay above six. Never discharge a child with a cleared bilirubin from specialist follow-up; instead, follow the second clock with platelets, spleen size, stiffness and oxygen saturation, and act on new portal hypertension. Prepare adolescents and their adult doctors for a second period of risk. And when parents ask whether their child will keep her liver, give them an honest number for today, and tell them that you will update it together at every visit.